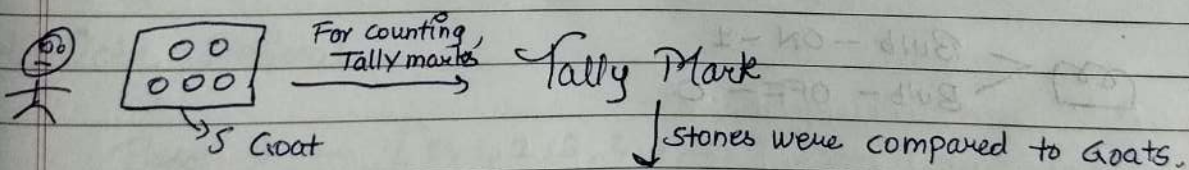
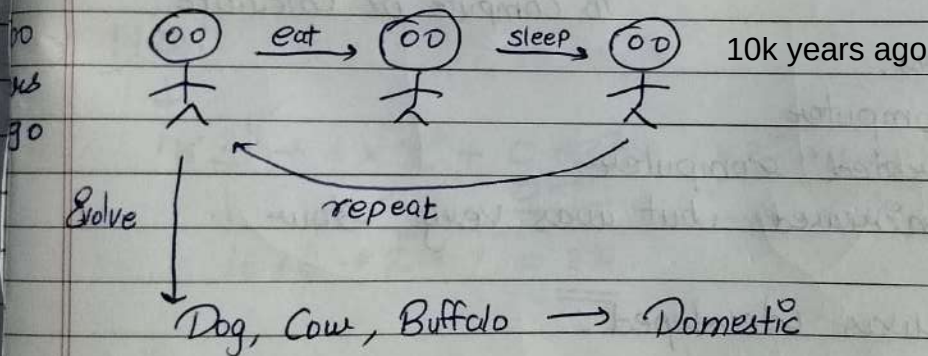
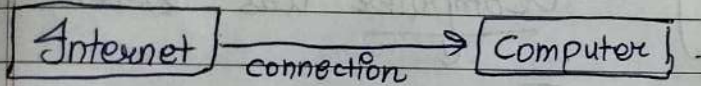


ULTIMATE WEB-DEV COURSE

* Lecture - 1 - Introduction

- 1) Respect the course and complete it.
- 2) Maintain consistency.
- 3) Make a fixed schedule.
- 4) Make notes.
- 5) Deploy code on GitHub.
- 6) Maintain accountability.
- 7) Your Resume should motivate you.
- 8) After live class, try to solve questions yourself.



- 1000 Goat $\rightarrow$ Now what?

Number System $\rightarrow$

Base 60

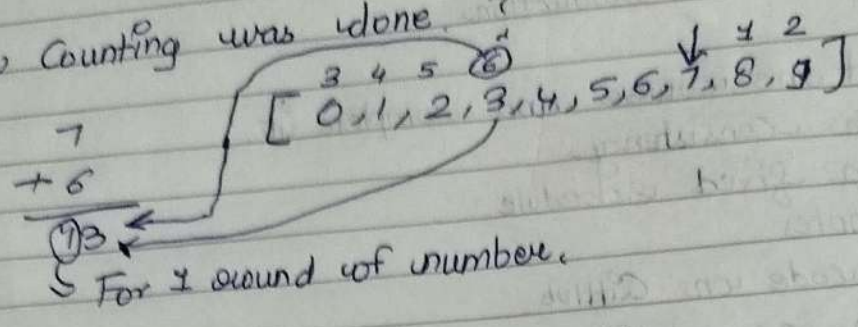
60 unique digits
Egypt number system

* Base 10 $\rightarrow$ 10 unique digits.

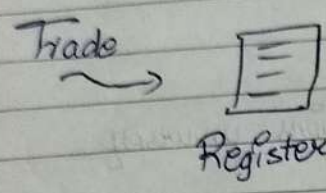
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]

- Developed from India.

How Counting was done?



1800
Century



Was difficult to calculate.

Register.

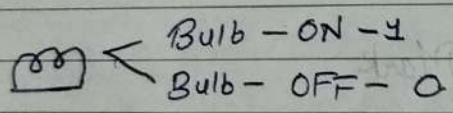
Computer was developed

To Compute or Calculate.

- Big Computers, size of one room
- Physical Computer.
- Human operated computer
- Accuracy increased but was very slow.

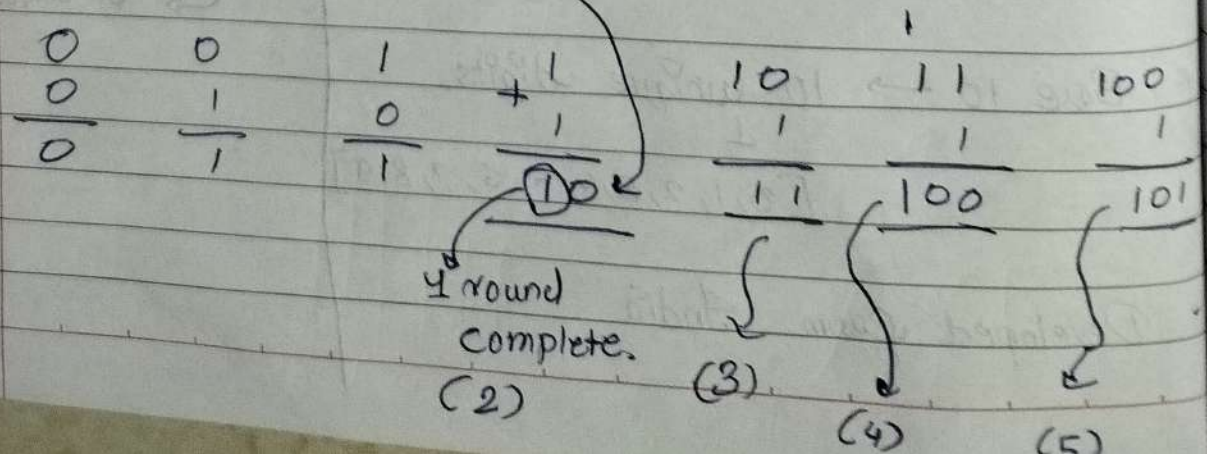
1950

Transistor was developed.



- Binary number system was developed.

Base 2 [0, 1]



* Decimal to Binary

1) 27

2	27	
2	13	1
2	6	1
2	3	0
2	1	1
0	0+1=1	

→ 11011

2) 57

2	57	
2	28	1
2	14	0
2	7	0
2	3	1
2	1	1
0	1	

→ 111001

* Binary to decimal

1) 11011

$$1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$16 + 8 + 0 + 2 + 1 = 27$$

* Octa-Decimal

Base - 8 → [0, 1, 2, 3, 4, 5, 6, 7]

8	37	
8	4	5
	0	4

→ (45)₈

$$4 \times 8^1 + 5 \times 8^0$$

$$32 + 5 = 37$$

* Hexa-decimal

Base 16 → [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F]

16 unique characters

1) 38

16	38	
16	2	6
	0	2

↑ (26)₁₆

$$(26) \rightarrow 2 \times 16^1 + 6 \times 16^0$$

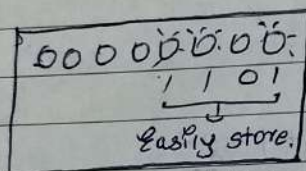
$$\downarrow$$

$$32 + 6 = 38$$

* Transistor

↓
With help of transistors & number systems,
computers were optimized.

Binary number system made it easy to store numbers.



13 store ?

1101

31 store ?

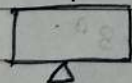
11111



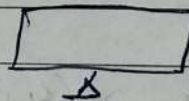
That bulbs were kept on just off!

- RAM size increased.
- Transistors increased rapidly, computer system came in 1950's

1950



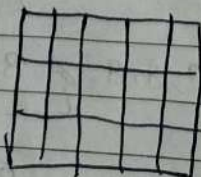
How to transfer data ?



- Tape or sheet was used.

rows & columns

14
↓
1110



If 1 then hole, If 0 then no hole

Interaction were

Sheet was taken to h

Punching Task